

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0260

Ansonia 138 kV
June 15, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0260. The Interconnection Customer proposes a 199.1 MW solar photovoltaic facility interconnecting at the Ansonia 138 kV substation in the New Jersey area. The request was received on August 10, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 204 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,632,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: White Rock-Nike / Magnolia 1708 138kV Line
Contingency: Shredder / Roxboro 138kV line
Loading: 180.5% (Rating: 217 MVA)
DFAX: 5.0%
Pre-project loading: 175.9%
2. Facility: Ansonia / Fanwood Substation 138kV
Contingency: Meadow Rd / Hoist 138kV
Loading: 113.0% (Rating: 242 MVA)
DFAX: 21.4%
Pre-project loading: 95.4%
3. Facility: Pine Tree Acres / Crawfordsville Power Plant 138kV L
Contingency: Magnolia 1708 / White Rock-Nike 138kV line
Loading: 100.1% (Rating: 432 MVA)
DFAX: 6.1%
Pre-project loading: 97.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
White Rock-Nike / Magnolia 1708 138kV Line	Rebuild line with higher-capacity conductor	\$1,436,000	12% (\$166,000)
Fanwood / Ansonia 138kV	Rebuild line with higher-capacity conductor	\$2,245,000	100% (\$2,245,000)
Crawfordsville Power Plant / Pine Tree Acres 138kV L	Reconductor line	\$513,000	43% (\$221,000)
	Total allocated to project		\$2,632,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.